# PrepByEKS

TAPIR Point Tracker v2

## User Manual

Institut Sukan Negara Malaysia
BPS Division — Sports Science Centre
Edin Kardin Suwarganda

# PrepByEKS — TAPIR Point Tracker v2

## User Manual

### Before You Start

Make sure you have a video file ready (MP4, AVI, MOV, or MKV). The app tracks points you place on a video and converts their movement into real-world distance, velocity, and acceleration — but only after you calibrate it using known measurements from the scene.

### 1. Launch the App

Double-click `launch.bat` inside the app folder.

A window opens titled **PrepByEKS | TAPIR Point Tracker v2**. At the bottom is a green status bar that reads: *Open a video to begin.*

The window has three tabs:

- **Setup**

## 4. Place Tracking Points (Setup Tab)

The Setup tab shows the video frame on the left and a sidebar on the right.

### *Placing points*

Left-click anywhere on the frame to place a point. Each point is automatically labelled P1, P2, P3 and so on. A coloured dot appears on the frame and the name appears in the list on the right. You can place up to 30 points.

### *Renaming a point*

Double-click the point's row in the list, or select it and click **Rename**. Type the new name and press Enter or click OK.

### *Changing a point's colour*

Select the row in the list and click **Color**. Pick a colour from the colour picker and click OK.

### *Removing a point*

**Right-click** on the canvas near a point to remove the nearest one. Or select the row in the list and click **Remove**.

### *Clearing all points*

Click **Clear All** to remove every point at once.

### *Using the magnifier for precise placement*

Click **Magnify** in the top-right of the points panel. The button turns teal and reads *Magnify ON*.

Move your mouse over the frame. A 180×180 px loupe appears next to the cursor and follows it. It shows:

- A magnified view (~3.6×) of the area around the cursor
- A gap crosshair with a circle at the exact centre
- Pixel coordinates (x, y) at the bottom of the loupe
- Already-placed tracking points drawn in their colours inside the loupe

Click to place a point exactly at the crosshair centre. Click **Magnify ON** again to turn the loupe off.

The loupe flips to the other side of the cursor when you approach an edge so it never covers the area you are clicking.

To remove a connection, select it in the list and click **Remove Selected**. To remove all, click **Clear All**.

## 6. Run Tracking

Trac Tab) complete — switch to Replay ta545 Tj T* ET Q Q q 1 0 0 1 57.0235294

Click **Run Tracking** in the top bar.

**On the first run only**, the app downloads pretrained model weights from the internet (about 200 MB). This takes a minute depending on your connection. Wait for the progress bar to complete.

When tracking finishes:

- The status bar says *Tracking complete — switch to Replay tab.*



Tracking complete — switch to Replay ta590 Ti T* ET Q Q q 1 0 0 1 57.02362 695cluQ q 1 0C94 92tch thracSm

Switch to the **Calibrate** tab. The frame is on the left. The right panel is scrollable — use the scrollbar or mouse wheel to reach all sections.

You can add multiple calibrations of either type. Different tracked points can use different calibrations.

---

### *9A. Linear Calibration — 2 points*

Use this when you know the straight-line distance between two visible points in the frame. Examples: two lane markings, the ends of a measuring tape, the width of a known object.

1. Select **Linear (2 pts)** mode at the top of the sidebar.
2. Click **Magnify** for precision if needed. The loupe shows existing calibration marks in colour.
3. Click the **first point** on the frame. A dot labelled **1** appears.
4. Click the **second point**. A dot labelled **2** appears and a line connects them. Right-click to undo the last dot.
5. In the **LINEAR CALIBRATION** section, type a **Name** (e.g. *Lane*).
6. Type the **Distance (m)** — the real-world distance in metres between your two points (e.g. *1.00*).
7. Click **+ Add Linear Calibration**.

The calibration appears in the list and the status bar shows the pixel-per-metre ratio.

Repeat for more linear calibrations if needed.

---

### *9B. Field Calibration — 4 points (perspective-corrected)*

Use this when you have a known rectangular area visible in the frame — for example, a sports court, a mat, a marked area on the floor, or any flat rectangle with known dimensions.

This method uses a perspective transform (homography) to correct for camera angle. Points anywhere inside the quadrilateral are correctly scaled even if the camera is not overhead.

1. Select **Field (4 pts)** mode.
2. Click **Magnify** for precision if needed.
3. Click the **four corners** of the known rectangle in this exact order:

**Top-Left** → **Top-Right** → **Bottom-Right** → **Bottom-Left**

A labelled dot (TL, TR, BR, BL) appears at each click. Right-click to undo the last dot.

1. In the **FIELD CALIBRATION** section, type a **Name** (e.g. *Court*).
2. Type **W (m)** — the real-world width of the rectangle in metres.
3. Type **H (m)** — the real-world depth (length) of the rectangle in metres.
4. Click **+ Add Field Calibration**.

A yellow quadrilateral appears on the canvas showing the calibrated region.

> *Use the largest rectangle you can identify for best accuracy. The corners do not need to look like a rectangle on screen — the app corrects for the camera angle.*

---

### *9C. Removing a calibration*

Select a calibration in its list and click the **Remove** button beneath that list.

---

## 10. Assign Calibrations to Points (Calibrate Tab)

Adding calibrations alone is not enough — you must assign each tracked point to the calibration it should use.

Scroll down to **ASSIGN TO POINTS** at the bottom of the sidebar.

1. Choose a tracked point from the **Point** dropdown (e.g. *P1*).
2. Choose a calibration from the **Calib** dropdown. Linear calibrations are listed as *L: name*, field calibrations as *F: name*.
3. Click **Assign**.

The assignment appears in the list, for example: `P1 -> [L] Lane`.

Repeat for every point you want kinematics for. Points with no assignment are skipped.

Each point can use a different calibration — P1 can use a linear calibration and P2 can use a field calibration.

To remove an assignment, select it in the list and click **Clear Assignment**.

---

## 11. Compute Kinematics and Export (Replay Tab)

Switch back to the **Replay** tab.

In the **Kinematics** section at the bottom right:

1. Optionally adjust **Win** and **Poly** for the derivative smoothing. These are independent from the trajectory smoothing controls on the left.
2. Click **Compute & Export**.

A confirmation dialog shows the saved file path. The kinematics CSV is saved to the **output** folder.

### *Kinematics CSV columns*

| Column | Unit | Description |
|---|---|---|
| frame | — | Frame number |
| time_s | s | Time in seconds |
| point_id | — | Point number |
| label | — | Point name |
| calibration | — | Calibration used |
| world_x_m | m | Real-world horizontal position |
| world_y_m | m | Real-world vertical position |
| vel_x_ms | m/s | Horizontal velocity |
| vel_y_ms | m/s | Vertical velocity |
| speed_ms | m/s | Resultant speed |
| accel_ms2 | m/s² | Acceleration (rate of change of speed) |
| distance_m | m | Cumulative distance travelled |

---

## 12. Export Annotated Video and Pixel Tracks

Click **Export** in the top bar.

This saves two files to the **output** folder:

| File | Contents |
|---|---|
| `*_tapir_tracked.mp4` | Annotated video with points, trails, connections, and info overlay |
| `*_tracks_*.csv` | Raw pixel coordinates (x, y) for every point at every frame |

## 13. Output Folder

```
Tapir-Tracker\output\
<videoname>_tapir_tracked.mp4
<videoname>_tracks_<timestamp>.csv
<videoname>_kinematics_<timestamp>.csv
```

## 14. Config Settings (config.py)

Open `config.py` in any text editor to adjust these settings:

| Setting | Default | Effect |
|---|---|---|
| `DISPLAY_MAX_W` | 960 | Display width — reduce if you get memory errors |
| `DISPLAY_MAX_H` | 540 | Display height — reduce if you get memory errors |
| `OCCLUSION_THRESHOLD` | 0.3 | Lower = more points stay visible through occlusion |
| `COTRACKER_WINDOW_SIZE` | 8 | Higher = better tracking of slow movement, slower to run |
| `SMOOTH_WINDOW_DEFAULT` | 11 | Default smoothing window |
| `SMOOTH_POLYORDER_DEFAULT` | 3 | Default polynomial order |
| `MAX_POINTS` | 30 | Maximum tracked points |
| `TRAIL_LENGTH` | 60 | Frames of trail history shown |

## 15. Troubleshooting

| Problem | What to do |
|---|---|
| App does not start | Use `launch.bat`, not `python main.py` directly |
| Out of memory on video load | Reduce `DISPLAY_MAX_W` and `DISPLAY_MAX_H` in `config.py` |
| Tracking is very slow | Reduce `COTRACKER_WINDOW_SIZE` to 4 in `config.py` |
| Points lose track frequently | Reduce `OCCLUSION_THRESHOLD` to 0.2; increase `COTRACKER_WINDOW_SIZE` to 16 |
| App hangs at "Loading model" on first run | Waiting for ~200 MB download — check internet connection |

| Problem | What to do |
|---|---|
| Kinematics CSV is all zeros for a point | That point has no calibration assigned — go to Calibrate tab and assign one |
| Wrong scale in kinematics output | Check W and H values are in **metres**, not centimetres |
| Field calibration looks distorted | Re-do corners in the correct order: TL → TR → BR → BL |
| Missing packages | Run `install_deps.bat` from the app folder |

## 16. Full Workflow at a Glance

```
1. Double-click launch.bat
2. Click Open Video
3. Use A / D keys to find the best frame
4. Click Magnify → place tracking points with precision (left-click)
5. Rename and colour points as needed
6. Add line connections if needed (optional)
7. Click Run Tracking — wait for model download on first use
8. Review tracking in Replay tab
9. Adjust smoothing (Win and Poly) → click Apply
10. Draw annotations on key frames if needed (optional)
11. Switch to Calibrate tab
12. Select Linear or Field mode
13. Click Magnify → place calibration points precisely
14. Enter real-world measurements → click Add
15. Repeat for additional calibrations if needed
16. Scroll to Assign to Points → assign each point to a calibration
17. Switch to Replay tab
18. Click Compute & Export → kinematics CSV saved to output\
19. Click Export → annotated video and pixel CSV saved to output\
```

*PrepByEKS · Institut Sukan Negara Malaysia · BPS Division*

*Edin Kardin Suwarganda*